

ISLAMIC UNIVERSITY OF TECHNOLOGY



COMPUTER NETWORKS LAB
CSE 4512

Lab 2 Report

Submitted by:

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Section: 1B

1. Task Description

Iutians Assemble headquarters consists of the following departments:

- Engineering (Floor 1)
- Marketing (Floor 2)
- Sales (Floor 3)
- Administration (Floor 4)
- Management (Floor 5)

All departments share the addressing format:

$$192.(i).(i + 3\%14)$$

Where **i = 22 (student ID 220041122)**.

The number of devices per department is determined as:

$$E(i) = (7i + 3) \bmod 41 + 10 \quad (\text{Engineering})$$

$$M(i) = (5i + 11) \bmod 26 + 5 \quad (\text{Marketing})$$

$$S(i) = (11i + 7) \bmod 36 + 5 \quad (\text{Sales})$$

$$A(i) = (3i + 17) \bmod 18 + 3 \quad (\text{Administration})$$

$$G(i) = (13i + 1) \bmod 9 + 2 \quad (\text{Management})$$

The task goal is to design a scalable network where all departments can communicate.

2. Network Topology

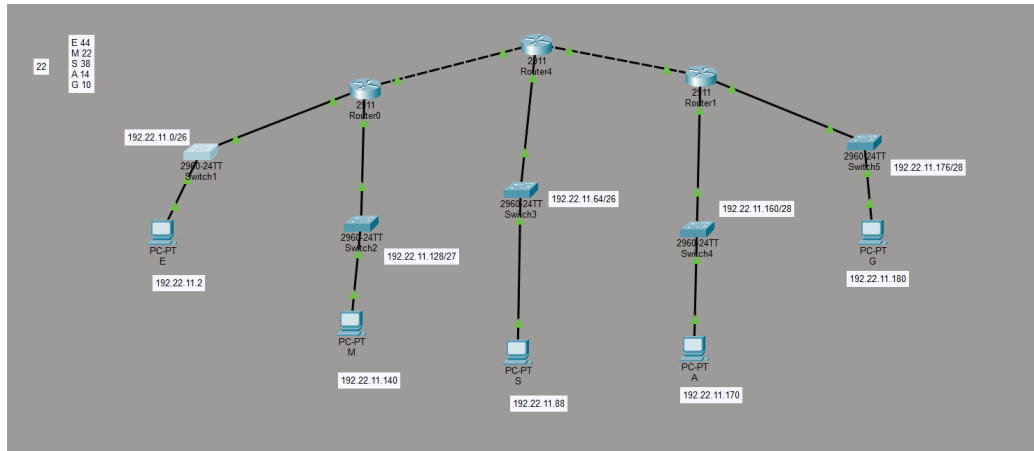


Figure 1: Subnetted Multi-Router Network Topology

Five subnetworks were created and assigned to routers, switches and end devices. Each department received an independent subnet to maintain scalability, management ease and secure segmentation.

3. Subnetting Worksheet

Department	Devices	Network Address	Mask	Usable Host Range
Engineering	$E(i)$	192.22.11.0	/26	192.22.11.1 – 192.22.11.62
Sales	$S(i)$	192.22.11.64	/26	192.22.11.65 – 192.22.11.126
Marketing	$M(i)$	192.22.11.128	/27	192.22.11.129 – 192.22.11.158
Administration	$A(i)$	192.22.11.160	/28	192.22.11.161 – 192.22.11.174
Management	$G(i)$	192.22.11.176	/28	192.22.11.177 – 192.22.11.190

Each subnet was assigned to a router interface and connected to switches hosting PCs.

4. Procedure

1. Network Construction

- 3 Routers, 5 Switches, 5 PCs were placed in Cisco Packet Tracer.
- Straight-through cables connected PC–Switch and Switch–Router.
- Routers were interconnected for inter-department communication.

2. Address Assignment

- Subnets were allocated as shown in the worksheet.
- Default gateway = Router interface IP of respective subnet.

3. Routing Configuration

- RIP routing was implemented on all routers.

4. Connectivity Verification

- Ping tests were executed across all departments. Successful reply would confirm full connectivity.

5. Results & Observations

All PCs could not reach each other across departments because of VLAN configuration issues. However routing table was successfully made. Subnet utilization was also done properly.

6. Challenges Faced

- Wrong subnet mask initially caused routing failure.
- Some PCs could not ping for not following subnet mask IP configuration and wrong gateway assignment.
- VLAN configuration was not setup.